Venkata Lakshmi Parimala Pasupuleti

Full Stack AI Engineer · Forward Deployed · Agentic AI · LangGraph · RAG · LLMOps · MLOps · Cloud-Native (AWS · Azure · GCP)

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SUMMARY

Full-stack AI engineer with 9+ years building enterprise production systems and two years deploying agentic AI solutions end-to-end. Built and deployed **Leasel** - a production multi-agent LangGraph system with RAG pipelines, hybrid search, HITL escalation, audit logging, and responsible AI compliance - directly for client operations. Built **AgentProbe** - open-source LLMOps and MLOps toolkit with offline evals, tracing, feedback loops, and CI regression harness. Deep expertise across full-stack engineering (Python, Java, JavaScript/TypeScript, Node.js, React, Angular, Spring Boot, .NET), agentic AI and Generative AI (LangChain, LangGraph, OpenAI, Anthropic, Azure OpenAI, Gemini, Llama, Hugging Face, model fine-tuning), cloud-native architecture (AWS, Azure, GCP, multi-cloud, Kubernetes, Docker, Terraform, IaC), DevSecOps (CI/CD, GitOps, Secure SDLC, observability), and data engineering (ETL/ELT, Apache Kafka, data lakes, real-time streaming). At Entrata (9+ years, 100K+ users), translated complex enterprise requirements into production software across a regulated multi-tenant platform. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Agentic AI · AI Agents · Multi-Agent Orchestration | LangChain · LangGraph · RAG Pipelines | Generative AI · LLMOps · MLOps | Pinecone · Weaviate · ChromaDB · Vector Databases | Model Fine-Tuning · Hugging Face · Prompt Engineering | OpenAI · Anthropic · Azure OpenAI · Gemini · Llama | Python · Java · JavaScript · TypeScript · Node.js | React · Angular · Spring Boot · .NET | Full-Stack · Microservices · REST APIs | Distributed Systems · Event-Driven Architecture | AWS · Azure · GCP · Multi-Cloud · Cloud-Native | Docker · Kubernetes · Serverless | CI/CD · GitOps · Terraform · IaC | Observability · Monitoring · OpenTelemetry | Secure SDLC · OWASP LLM · Secure AI | ETL/ELT · Data Pipelines · Apache Kafka | Enterprise Architecture · System Integrations | Forward Deployed Engineering

EXPERIENCE

AI/ML Engineer (Volunteer)

Sep 2026 - Present

Blue Horse Digital — Virginia, USA

- Building ML models, AI-powered features, and data pipelines end-to-end; integrating AI components with backend APIs and surfacing insights into application-facing reporting layers for business stakeholders.

AI/ML Engineer Intern (Forward Deployed)

May 2025 - Aug 2025

Magizhs Tech Solution

- Designed, built, and deployed a production agentic AI system directly for client operations - framed business requirements into an architecture, prototyped with stakeholders, and shipped a LangGraph stateful multi-agent workflow (parse → plan → retrieve → validate → route) with full RAG pipeline, FastAPI serving, and enterprise system integration via MCP server.
- Engineered validation guardrails enforcing hard compliance boundaries, structured audit logging, and secure AI practices throughout; deployed with CI/CD pipeline and defined API contracts connecting agent outputs to downstream enterprise systems.

Senior Software Engineer

Sep 2018 - Aug 2024

Entrata - PropTech / Real Estate SaaS (100K+ monthly active users, enterprise multi-tenant)

- **Shipped 20+ full-stack production modules** (Python, PHP, Java, JavaScript, PostgreSQL) across 120+ Agile sprints - microservices architecture (Docker, 50% scalability improvement), event-driven processing (RabbitMQ async pipelines with retry/idempotency), and REST API development for a regulated multi-tenant enterprise platform.
- **Owned cloud migration and data pipeline infrastructure** - PHI-compliant AWS S3 migration (Libsodium encryption, 100% coverage, zero security incidents); HUD compliance ETL pipeline (federal rows → 180,000 domain records) powering eligibility and rent calculations for 100K+ housing units; optimized 200+ SQL queries (30% gain).
- **Built observability and DevSecOps tooling** - internal debugging tools (40% debugging time reduction); CI/CD automation eliminating 80% of manual workload; owned production incident response, root-cause analysis, and postmortems; maintained bug count below 10 for 6+ consecutive months.
- Cross-functional stakeholder delivery across product, operations, compliance, and business teams; mentored 5+ engineers; 300+ code reviews; Zero compliance violations across 7 years.

Associate Software Engineer

Jul 2017 - Sep 2018

Entrata India, Pune

- Built backend modules (Python, PHP, Java, PostgreSQL); led AWS S3 cloud migration; delivered large-scale ETL and data reconciliation workflows (1.6M records); designed and migrated document storage database architecture.

PROJECTS

Leasel - Production Agentic AI Platform (Multi-Agent · RAG · Responsible AI) 2026 | Python, LangGraph (8-agent), FastAPI, ChromaDB, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built and deployed a three-surface enterprise AI platform: **multi-agent LangGraph pipeline** (8 agents: Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → HITL Escalation) with durable state management and conditional routing; **RAG pipeline** with hybrid search (TF-IDF + semantic reranking), citation grounding, and vector database retrieval; **self-service interface** with hard safety boundaries enforced in code. Secure AI throughout: audit logging on every agent decision, OWASP LLM guardrails, access-controlled retrieval, prompt injection defense.

- CI/CD regression gate; 93.6% precision and 87.9% F1 across 1,240 test cases; deployed to production with monitoring and observability.

AgentProbe - LLMOps & MLOps Evaluation Toolkit 2025 | Python, FastAPI, LangGraph, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Open-source LLMOps and MLOps eval toolkit: offline evals with golden datasets and LLM-as-judge scoring; agent tracing and structured observability; CI-integrated regression harness; 5-metric eval engine (accuracy, hallucination rate, retrieval quality, completeness, compliance); multi-provider routing (OpenAI, Anthropic, Azure OpenAI, Gemini, Groq, Ollama/Llama). Model fine-tuning evaluation support via Hugging Face integration.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia | GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017